The Monitor tab and the job list

What runs on a screener tool by itself, when, and how to see and change it. Written from `src/scheduler.js`, `src/routes/monitor.js` and `src/pipeline.js` (`getScanStatus`).

Related: what each scan does (`docs/DATA-PIPELINE.md`), what the captures freeze (`docs/REGISTERS.md`), the settings these jobs read (`docs/SETTINGS.md`).

The tab

Two sections:

Section	Shows
Scheduled Jobs	one row per job: name, schedule, on/off, when it last ran, how long it took, ok or the error. Per row: **toggle**, **edit schedule**, **reset to default**
Last Pipeline Run	the last scan: when, rows found, whether one is running now, its error, and its report (stage by stage) plus when the live cards were last refreshed. If the tool is paused, when and why

Both come from one call, `GET /api/monitor`:

```
{
  "pipeline": { "paused", "pausedAt", "pausedReason", "lastRun", "lastRowCount",
                "running", "error", "lastReport", "lastRefresh" },
  "jobs": [ { "jobId", "name", "defaultSchedule", "schedule", "timezone",
              "enabled", "lastRunAt", "lastStatus", "lastError", "lastDuration" } ]
}
```

The jobs

All times are New York time (`America/New_York`). `1-5` = Monday to Friday.

Job	Default schedule	What it does
Discovery — Pre-Market (04:00–09:00)	`*/30 4-8 * * 1-5`	a full scan every 30 min
Discovery — Open (09:00–10:00)	`*/5 9 * * 1-5`	a full scan every 5 min
Discovery — Session (10:00–16:00)	`*/15 10-15 * * 1-5`	a full scan every 15 min
Discovery — Close (16:00)	`0 16 * * 1-5`	one last full scan
Refresh — Live Card Data (04:00–16:00)	`*/5 4-16 * * 1-5`	re-quotes the cards already found; adds nothing new
Shortlist Auto-Rule HH:MM	one minute before R1	stars the top cards by score (`docs/SETTINGS.md`); needs the scorer (D-D1)
R1 Capture HH:MM	the tool’s capture time (default 09:36)	freezes the photo (R1)
R2 Snapshot 9:26–9:56 AM	`26,31,36,41,46,51,56 9 * * 1-5`	an open-range snapshot every 5 min
R2 Snapshot 10:01 AM	`1 10 * * 1-5`	the eighth and last R2 snapshot
R3 EOD Capture 4:05 PM	`5 16 * * 1-5`	the day’s outcomes (R3) and the joined R4
Scorer Auto-Train 4:20 PM	`20 16 * * 1-5`	retrains the tool’s scorer on all R4 history; skipped when there are not enough rows or the scorer is off
Midnight r0 Flush	`0 0 * * *` (every day)	empties today’s live list (r0)
Apply tool schedule changes 4:50 PM	`50 16 * * 1-5` , **T1 only**	restarts the tools whose capture time was changed during the session (`docs/TOOL-LIFECYCLE.md`)

What a full scan and a refresh do: `docs/DATA-PIPELINE.md` .

Rules around the jobs

- **A paused tool skips scans.** A full scan or refresh on a paused tool returns at once and says it is paused; the captures still run on what is already in the list.

- **No overlap.** A scan that starts while another is running is skipped (`Scan already running`); a refresh during a scan is skipped too.
- **Errors are logged with the stack**, not just the message, so a job that fails every five minutes can be traced to a line.
- **R1 needs the screeners awake.** The capture time is set per tool so the photo is taken while that tool's screeners can have found something (D-S9 is the T6 case where it is not).

Changing a job

Action	Route	Effect
Toggle	`POST /api/monitor/jobs/:jobId/toggle`	on ↔ off; an off job keeps its row and schedule
Edit schedule	`POST /api/monitor/jobs/:jobId/schedule` body `{ "schedule": "<cron>" }`	refused unless the cron expression is valid; applies at once
Reset	`POST /api/monitor/jobs/:jobId/reset`	back to the default schedule

What is kept, and where

- **Kept across restarts** (table `scheduler_jobs` in the tool's database): each job's schedule and on/off.
- **Lost on restart** (memory only): last run, status, error, duration.
- **A job's id is its name, slugified** (`R1 Capture 09:36` → `r1-capture-09-36`). So when a tool's capture time changes, the R1 and Auto-Rule jobs get new names and new ids: they start from their defaults, and a custom schedule or "off" set on the old ids is dropped. Rows left behind by renamed jobs are deleted at startup.

Reading it

You see	It means
Discovery rows ok, `Last Pipeline Run` recent	the tool is scanning
Every row ok but no cards	screeners matched nothing; check the screener windows and the floor
`paused` with a reason	someone paused it (`docs/TOOL-LIFECYCLE.md`); resume on the Settings tab
Auto-Train “skipped” in the log	not enough R4 rows yet, or the tool has no scorer
Last run empty after a restart	normal: run history is memory only